

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.51)

Metric

$$ds^2 = -dt \otimes dt + \left(\frac{F(t)^2}{\epsilon r^2 + 1} \right) dr \otimes dr + V(t, z, r, \phi)^2 dz \otimes dz + r^2 F(t)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, z, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\frac{\sqrt{2}F(t)}{2\sqrt{\epsilon r^2 + 1}} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\frac{\sqrt{2}F(t)}{2\sqrt{\epsilon r^2 + 1}} \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2}V(t, z, r, \phi) \right) dz + \left(\frac{1}{2} i \sqrt{2}rF(t) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2}V(t, z, r, \phi) \right) dz + \left(-\frac{1}{2} i \sqrt{2}rF(t) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(\frac{i\sqrt{2}}{2rF(t)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(-\frac{i\sqrt{2}}{2rF(t)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z + \frac{i\sqrt{2}}{2rF(t)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z - \frac{i\sqrt{2}}{2rF(t)} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}(\epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) + \epsilon r^2 V(t, z, r, \phi) - \sqrt{\epsilon r^2 + 1} r V(t, z, r, \phi) \frac{\partial}{\partial t} F(t) - \sqrt{\epsilon r^2 + 1} r F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + r \frac{\partial}{\partial r} V(t, z, r, \phi) + V(t, z, r, \phi))}{4\sqrt{\epsilon r^2 + 1} r F(t) V(t, z, r, \phi)}$$

$$\sigma = -\frac{\sqrt{2}(\epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) - \epsilon r^2 V(t, z, r, \phi) + \sqrt{\epsilon r^2 + 1} r V(t, z, r, \phi) \frac{\partial}{\partial t} F(t) - \sqrt{\epsilon r^2 + 1} r F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + r \frac{\partial}{\partial r} V(t, z, r, \phi) - V(t, z, r, \phi))}{4\sqrt{\epsilon r^2 + 1} r F(t) V(t, z, r, \phi)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}(\epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) + \epsilon r^2 V(t, z, r, \phi) + \sqrt{\epsilon r^2 + 1} r V(t, z, r, \phi) \frac{\partial}{\partial t} F(t) + \sqrt{\epsilon r^2 + 1} r F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + r \frac{\partial}{\partial r} V(t, z, r, \phi) + V(t, z, r, \phi))}{4 \sqrt{\epsilon r^2 + 1} r F(t) V(t, z, r, \phi)}$$

$$\lambda = -\frac{\sqrt{2}(\epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) - \epsilon r^2 V(t, z, r, \phi) - \sqrt{\epsilon r^2 + 1} r V(t, z, r, \phi) \frac{\partial}{\partial t} F(t) + \sqrt{\epsilon r^2 + 1} r F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + r \frac{\partial}{\partial r} V(t, z, r, \phi) - V(t, z, r, \phi))}{4 \sqrt{\epsilon r^2 + 1} r F(t) V(t, z, r, \phi)}$$

$$\nu = 0$$

$$\alpha = \frac{i \sqrt{2} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)}$$

$$\beta = \frac{i \sqrt{2} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)}$$

$$\gamma = \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)}$$

Ricci Tensor Components

$$\Lambda = \frac{\epsilon r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi) - r^2 V(t, z, r, \phi) \frac{\partial}{\partial t} F(t)^2 - 2 r^2 F(t) V(t, z, r, \phi) \frac{\partial^2}{(\partial t)^2} F(t) - 2 r^2 F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) - r^2 F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi) + 2 \epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) + \epsilon r^2 V(t, z, r, \phi) + r^2 F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 r^2 F(t)^2 V(t, z, r, \phi)}$$

Weyl Tensor Components

$$\Psi_0 = \frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi) + V(t, z, r, \phi) \frac{\partial}{\partial t} F(t)^2 - F(t) V(t, z, r, \phi) \frac{\partial^2}{(\partial t)^2} F(t) - F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi) + \epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi) - 2 \sqrt{\epsilon r^2 + 1} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

$$\Psi_1 = - \frac{i r F(t) \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi) - i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi) - i \sqrt{\epsilon r^2 + 1} r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi) + i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Psi_2 = \frac{\epsilon r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi) - r^2 V(t, z, r, \phi) \frac{\partial}{\partial t} F(t)^2 + r^2 F(t) V(t, z, r, \phi) \frac{\partial^2}{(\partial t)^2} F(t) + r^2 F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) - r^2 F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi) - \epsilon r^3 \frac{\partial}{\partial r} V(t, z, r, \phi) + \epsilon r^2 V(t, z, r, \phi) + r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{12 r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Psi_3 = \frac{i r F(t) \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi) - i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi) + i \sqrt{\epsilon r^2 + 1} r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi) - i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Psi_4 = \frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi) + V(t, z, r, \phi) \frac{\partial}{\partial t} F(t)^2 - F(t) V(t, z, r, \phi) \frac{\partial^2}{(\partial t)^2} F(t) - F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) + F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi) + \epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi) + 2 \sqrt{\epsilon r^2 + 1} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"